

Adaptive Inference: Dynamic Efficiency Strategies for Large Language Models

Komal Niraula, Junjie Mai
kn2505, jm11065

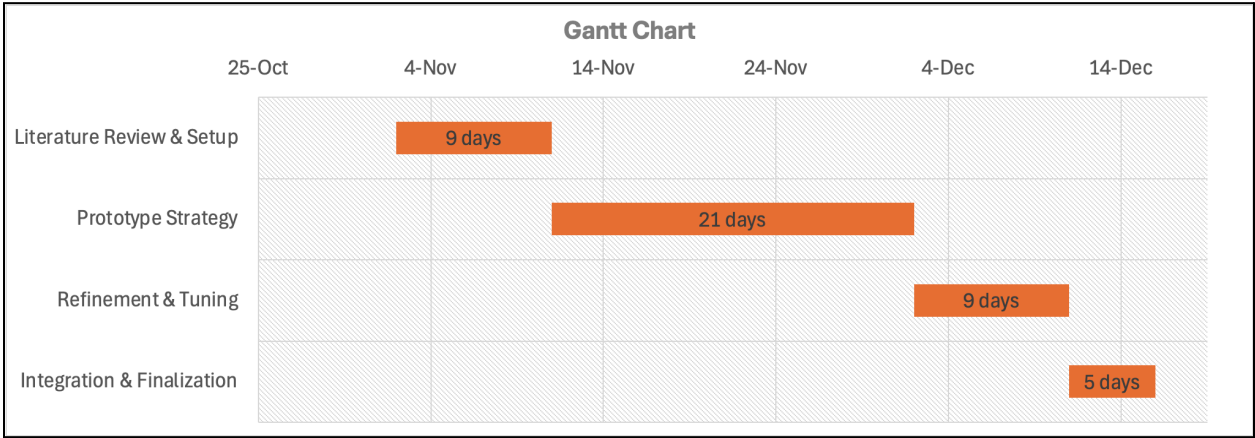
Large Language Models (LLMs) deliver exceptional ability across a wide range of tasks. However, they incur high inference cost because every input token traverses all transformer layers, regardless of task difficulty. Recent research suggests that not all layers are necessary for every instance (Fan et al., 2024). Our project uses GPT-2 as the primary model to explore adaptive inference. The study focuses on early-exit mechanisms that dynamically terminate computation, thereby reducing latency and energy use while maintaining accuracy.

We will focus on two primary categories: confidence based, and predictive early stopping methods. The confidence based approach relies on internal stability metrics like confidence from intermediate logits, terminating when the model’s confidence indicates that deeper layers are unlikely to alter predictions, as seen in CALM (Schuster et al. 2022) and LITE (Varshney et al., 2024). Predictive early exit methods such as AdaInfer (Fan et al., 2024) attach lightweight classifiers to intermediate layers to determine whether the current output sufficiently approximates the final prediction. To evaluate these approaches, the study will benchmark the strategies on a variety of classification and language-understanding datasets from Hugging Face, such as BoolQ for question answering and SST-2 for sentiment analysis. Performance will be assessed using accuracy and latency metrics. By evaluating multiple methods and analyzing their behavior, this project aims to contribute to effective adaptive inference. The results will help identify if early-exit mechanisms work across diverse tasks.

Individual responsibilities:

Stage	Junjie Mai	Komal Niraula
Literature Review & Setup	Collect and summarize related papers; document early-exit mechanisms and frameworks.	Prepare datasets, preprocessing pipelines, and baseline models for testing.
Prototype Phase	Collaborate on implementing early-exit strategies	
Refinement & Tuning	Analyze performance metrics, adjust thresholds, visualize trade-offs.	Improve model performance, latency reduction, or memory optimization.
Integration & Finalization	Prepare slides, charts, and presentation summary.	Write the main report and results discussion.

Gantt Chart:



References:

T. Schuster, A. Fisch, J. Gupta, M. Dehghani, D. Bahri, V. Q. Tran, Y. Tay, and D. Metzler, “Confident Adaptive Language Modeling,” Neural Information Processing Systems (NeurIPS), 2022. [Online]. Available: <https://arxiv.org/abs/2207.07061>

S. Fan, X. Jiang, X. Li, X. Meng, P. Han, S. Shang, A. Sun, Y. Wang, and Z. Wang, “Not All Layers of LLMs Are Necessary During Inference,” arXiv preprint arXiv:2403.02181, 2024. [Online]. Available: <https://arxiv.org/abs/2403.02181>

N. Varshney, A. Chatterjee, M. Parmar, and C. Baral, “Investigating Acceleration of LLaMA Inference by Enabling Intermediate Layer Decoding via Instruction Tuning with ‘LITE’,” Findings of the Association for Computational Linguistics: NAACL 2024, pp. 3656–3677, 2024. [Online]. Available: <https://aclanthology.org/2024.findings-naacl.232/>